TECHNICAL DOCUMENTATION

AI Corrector

Version 1.0.0

July 2, 2026

Contents

Changelog

- **02-07-2026**

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  Created initial documentation.

Base URL

Development	http://localhost:8000
UAT	https://<uat-host>:8000
Production	https://<prod-host>:8000

Authentication

When the API_KEY environment variable is set, all endpoints (except /, /health, /docs, /redoc, and /openapi.json) require the following request header:

Header Name	X-API-Key
Value	Your configured API key

Health Check

URL	{{BaseURL}}/health
Method	GET
Authorization	None

Response Fields

Field Name	Type	Note
status	string	Overall service status: "ok" or "degraded"
checks	object	See Health Checks

Health Checks

Field Name	Type	Note
vectordb	object	Azure AI Search connectivity status
vectordb.status	string	"ok" or "error"
vectordb.document_count	integer	Number of indexed documents
vectordb.error	string	Error message if status is "error"

Example Response

```
{
    "status": "ok",
    "checks": {
        "vectordb": {
            "status": "ok",
            "document_count": 150
        }
    }
}
```

Feed — Course Material Ingestion

Upload Files

URL	`{{BaseURL}}/feed`
Method	POST
Content-Type	`multipart/form-data`
Authorization	X-API-Key

Upload one or more course material files (PDF, PPT, PPTX, DOCX, TXT). Files are parsed, chunked, embedded, and stored in Azure AI Search.

Request Fields

Field Name	Type	Required	Note
course_code	string	Yes	Course code to associate materials with, e.g. `COMP6100`
files	file[]	Yes	One or more files (PDF, PPT, PPTX, DOCX, TXT)

Response Fields

Field Name	Type	Note
status	string	Always `"completed"`
results	array	Per-file result list; see Result Item
token_usage	object	Aggregated token usage; see Token Usage
Result Item		
results[].status	string	`"success"` or `"failed"`
results[].filename	string	Original filename
results[].total_chunks_saved	integer	Number of chunks stored
results[].token_usage	object	Per-file token usage

Field Name	Type	Note
results[].error	string	Error message (only when "failed")

Example Response


```
{
    "status": "completed",
    "results": [
        {
            "status": "success",
            "filename": "lecture1.pdf",
            "total_chunks_saved": 45,
            "token_usage": {
                "embedding_tokens": 2000,
                "embedding_cost_usd": 0.044,
                "vision_tokens": 500,
                "vision_cost_usd": 1.25,
                "total_cost_usd": 1.294
            }
        }
    ],
    "token_usage": {
        "embedding_tokens": 2000,
        "embedding_cost_usd": 0.044,
        "vision_tokens": 500,
        "vision_cost_usd": 1.25,
        "total_cost_usd": 1.294
    }
}
```


Upload from Single URL

URL	{{BaseURL}}/feed-url
Method	POST
Content-Type	application/json
Authorization	X-API-Key

Download a course material file from a URL and ingest it into the vector database.

Request Fields

Field Name	Type	Required	Note
url	string	Yes	Publicly accessible file URL
course_code	string	Yes	Course code to associate material with
token	string	No	Bearer token for protected URLs

Example Request

```json
{
    "url": "https://example.com/lecture1.pdf",
    "course_code": "COMP6100",
    "token": ""
}
```

Response Fields

Field Name	Type	Note
status	string	`"success"` or `"failed"`
message	string	Human-readable result message
total_chunks_saved	integer	Number of chunks stored
token_usage	object	See Token Usage

Example Response

```json
{
    "status": "success",
    "message": "'lecture1.pdf' inserted from URL",
    "total_chunks_saved": 45,
    "token_usage": {
        "embedding_tokens": 2000,
        "embedding_cost_usd": 0.044,
        "vision_tokens": 500,
        "vision_cost_usd": 1.25,
        "total_cost_usd": 1.294
    }
}
```

Upload from Multiple URLs

URL	`{{BaseURL}}/feed-urls`
Method	POST
Content-Type	`application/json`
Authorization	X-API-Key

Download and ingest multiple files concurrently. Individual failures do not fail the entire request.

Request Fields

Field Name	Type	Required	Note
urls	string[]	Yes	List of file URLs to ingest
course_code	string	Yes	Course code to associate materials with
token	string	No	Bearer token for protected URLs

Example Request

```
{
    "urls": [
        "https://example.com/lecture1.pdf",
        "https://example.com/lecture2.pptx"
    ],
    "course_code": "COMP6100",
    "token": ""
}
```

Response Fields

Field Name	Type	Note
status	string	Always `"completed"`
results	array	Per-file result list (same schema as `POST /feed`)
token_usage	object	Aggregated token usage; see Token Usage

Assessment — Student Answer Evaluation

Evaluate Single Student

URL	`{{BaseURL}}/assess`
Method	POST
Content-Type	`multipart/form-data`
Authorization	X-API-Key

Evaluate one student's answer against a question and rubric. When `use_key_answer` is `true`, the provided key answer is used as reference; otherwise the system retrieves relevant context from the vector database.

Request Fields

Field Name	Type	Required	Note
question	string	Yes	The exam question
student_answer	string	Yes	Student's answer text, or URL to document (Google Docs, PDF, web article)
student_answer_token	string	No	Bearer token for protected answer URLs
rubric	string	No	Rubric as a JSON array string; see Rubric Item
course_code	string	No	Course code for vector DB context (used when `use_key_answer=false`)

Field Name	Type	Required	Note
use_key_answer	boolean	No	`true`: use `key_answer_text` / file as reference. `false`: retrieve from vector DB. Default: `true`
key_answer_text	string	No	Reference answer text
key_answer_file	file	No	Reference answer file (PDF, PPT, PPTX, DOCX, TXT)

Example Request

```
{
    "question": "Jelaskan konsep machine learning!",
    "student_answer": "Machine learning adalah subset dari AI yang ...",
    "rubric": "[{\"minScore\":0,\"maxScore\":64,\"proficiency\":\"Poor\",\"
       ↪ criteria\":\"Unable to explain the concept\"},{\"minScore\":85,\"
       ↪ maxScore\":100,\"proficiency\":\"Excellent\",\"criteria\":\"Explains
       ↪  concept with clear examples\"}]",
    "use_key_answer": true,
    "key_answer_text": "Machine learning adalah cabang dari kecerdasan buatan
       ↪ ..."
}
```

Response Fields

Field Name	Type	Note
status	string	`"success"` or `"error"`
student_answer	string	Resolved plain text of the student's answer
retrieved_sources	array	Vector DB chunks used as context; see Retrieved Source
evaluation	object	AI evaluation result; see Evaluation Result
token_usage	object	See Token Usage

Example Response

```
{
    "status": "success",
    "student_answer": "Machine learning adalah subset dari AI ...",
    "retrieved_sources": [],
    "evaluation": {
        "reasoning": "Jawaban mencakup definisi dan konsep dasar dengan baik."
           ↪ ,
        "score": 85.5,
        "confidence": 92,
        "feedback": "Tambahkan contoh aplikasi praktis untuk skor lebih tinggi
           ↪ .",
        "sources": []
    },
    "token_usage": {
```

```
        "embedding_tokens": 1500,
        "embedding_cost_usd": 0.033,
        "vision_tokens": 0,
        "vision_cost_usd": 0.0,
        "completion_input_tokens": 2000,
        "completion_output_tokens": 500,
        "completion_cost_usd": 3.25,
        "total_cost_usd": 3.283
    }
}
```

Evaluate Batch — Single Question

URL	`{{BaseURL}}/assess-batch`
Method	POST
Content-Type	`application/json`
Authorization	X-API-Key

Evaluate multiple students for a single question in parallel. Individual student failures do not fail the entire request.

Request Fields

Field Name	Type	Required	Note
question	string	Yes	The exam question
rubric	array	No	Array of Rubric Item objects; see Rubric Item
course_code	string	No	Course code for vector DB context
use_key_answer	boolean	No	Use key answer or vector DB context. Default: `false`
key_answer	string	No	Reference answer text
students	array	Yes	Array of Student Answer objects; see Student Answer

Example Request

```
{
    "question": "Jelaskan apa yang dimaksud dengan machine learning!",
    "rubric": [
        {
            "minScore": 0,
            "maxScore": 64,
            "proficiency": "Poor",
            "criteria": "Unable to explain the concept"
        },
        {
            "minScore": 85,
```

```json
            "maxScore": 100,
            "proficiency": "Excellent",
            "criteria": "Explains concept with clear and practical examples"
        }
    ],
    "course_code": "COMP6100",
    "use_key_answer": false,
    "key_answer": "",
    "students": [
        {
            "student_id": "2501001234",
            "answer": "Machine learning adalah subset dari AI ...",
            "token": null
        },
        {
            "student_id": "2501001235",
            "answer": "https://docs.google.com/document/d/ABC123/edit",
            "token": null
        }
    ]
}
```

Response Fields

Field Name	Type	Note
status	string	`"success"`
retrieved_sources	array	Shared vector DB context chunks; see Retrieved Source
results	array	Per-student result list
results[].student_id	string	Student identifier
results[].status	string	`"success"` or `"error"`
results[].evaluation	object	AI evaluation; see Evaluation Result (null on error)
results[].student_answer	string	Resolved student answer text (null on error)
results[].error	string	Error message (only when `"error"`)
token_usage	object	Aggregated token usage; see Token Usage

Evaluate Batch — Multiple Questions

URL	`{{BaseURL}}/assess-batch-multi`
Method	POST
Content-Type	`application/json`
Authorization	X-API-Key

Evaluate multiple questions concurrently, each with its own set of students, rubric, and context source. The request body is an array of Batch Assess Request objects.

Request Fields

The request body is a JSON **array**. Each element is a `BatchAssessRequest` with the same fields as `POST /assess-batch`.

Example Request

```
[
    {
        "question": "Jelaskan machine learning!",
        "rubric": [],
        "course_code": "COMP6100",
        "use_key_answer": false,
        "key_answer": "",
        "students": [
            { "student_id": "2501001234", "answer": "...", "token": null }
        ]
    },
    {
        "question": "Jelaskan deep learning!",
        "rubric": [],
        "course_code": "COMP6100",
        "use_key_answer": true,
        "key_answer": "Deep learning adalah subset dari machine learning ...",
        "students": [
            { "student_id": "2501001234", "answer": "...", "token": null }
        ]
    }
]
```

Response Fields

Field Name	Type	Note
status	string	`"success"`
results	array	Per-question result list
results[].question	string	The original question text
results[].retrieved_sources	array	Vector DB context for this question
results[].results	array	Per-student results (same schema as `/assess-batch`)
token_usage	object	Aggregated token usage across all questions; see Token Usage

Shared Schemas

Rubric Item

Field Name	Type	Note
minScore	number	Minimum score threshold for this level
maxScore	number	Maximum score threshold for this level
proficiency	string	Proficiency label, e.g. `"Poor"`, `"Good"`, `"Excellent"`
criteria	string	Human-readable description of what earns this level

Student Answer

Field Name	Type	Note
student_id	string	Unique student identifier
answer	string	Answer text or URL to a document / Google Docs link
token	string	Bearer token for protected answer URLs (optional)

Evaluation Result

Field Name	Type	Note
reasoning	string	AI's step-by-step scoring rationale
score	number	Numeric score (range determined by rubric)
confidence	integer	AI confidence in the score (0–100)
feedback	string	Constructive feedback for the student
sources	array	Web sources cited by the AI (if web search was used)

Retrieved Source

Field Name	Type	Note
source	string	Source filename
page	integer	Page or slide number
content	string	Text content of the retrieved chunk

Token Usage

Field Name	Type	Note
embedding_tokens	integer	Tokens used for embedding
embedding_cost_usd	number	Embedding API cost (USD)
vision_tokens	integer	Tokens used by vision model
vision_cost_usd	number	Vision API cost (USD)
completion_input_tokens	integer	LLM prompt tokens (assessment only)

Field Name	Type	Note
completion_output_tokens	integer	LLM completion tokens (assessment only)
completion_cost_usd	number	LLM API cost (USD) (assessment only)
total_cost_usd	number	Total API cost (USD)